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Studierichting: Informatica
Oefeningenreeks 4A nr. 7.3

$$\begin{aligned} & \int \frac{-x^2\sqrt{x} - \sqrt[4]{x^3}}{x^{\frac{1}{3}}} dx \\ &= \int \frac{-x^2\sqrt{x}}{x^{\frac{1}{3}}} dx - \int \frac{\sqrt[4]{x^3}}{x^{\frac{1}{3}}} dx \\ &= \int \frac{-x^2 x^{\frac{1}{2}}}{x^{\frac{1}{3}}} dx - \int \frac{x^{\frac{3}{4}}}{x^{\frac{1}{3}}} dx \\ &= \int -x^{\frac{5}{2} - \frac{1}{3}} dx - \int x^{\frac{3}{4} - \frac{1}{3}} dx \\ &= \int -x^{\frac{13}{6}} dx - \int x^{\frac{5}{12}} dx \\ &= -\frac{6}{19} x^{\frac{19}{6}} - \frac{12}{17} x^{\frac{17}{12}} + C \end{aligned}$$

References